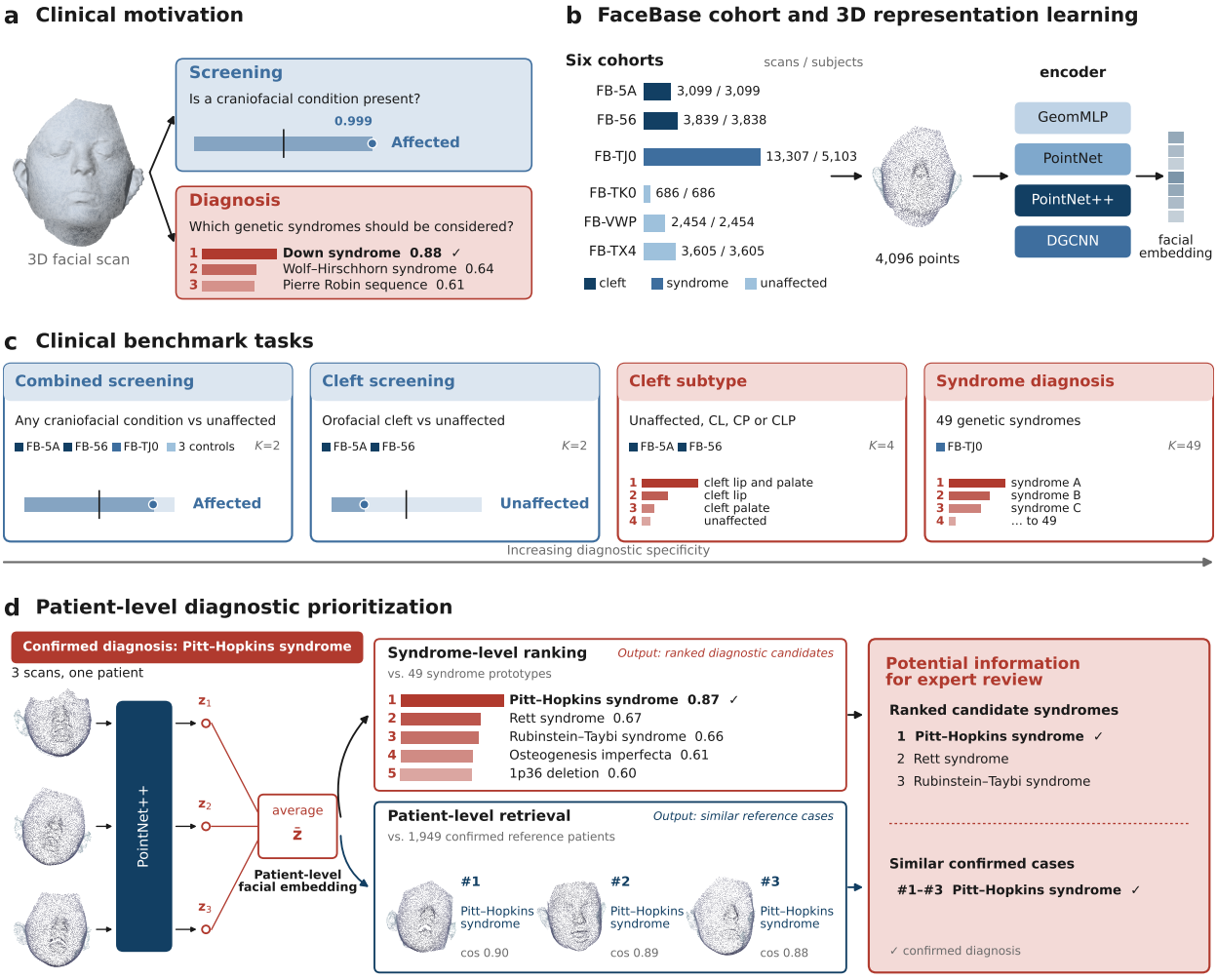

a Clinical motivation
3D facial scan
Screening
Is a craniofacial condition present?
0.999
Affected
Diagnosis
Which genetic syndromes should be considered?
1 Down syndrome 0.88 ✓
2 Wolf–Hirschhorn syndrome 0.64
3 Pierre Robin sequence 0.61
b FaceBase cohort and 3D representation learning
Six cohorts
scans / subjects
FB-5A 3,099 / 3,099
FB-56 3,839 / 3,838
FB-TJ0 13,307 / 5,103
FB-TK0 686 / 686
FB-VWP 2,454 / 2,454
FB-TX4 3,605 / 3,605
cleft
syndrome
unaffected
4,096 points
encoder
GeomMLP
PointNet
PointNet++
DGCNN
facial embedding
c Clinical benchmark tasks
Combined screening
Any craniofacial condition vs unaffected
FB-5A FB-56 FB-TJ0 3 controls K=2
Affected
Cleft screening
Orofacial cleft vs unaffected
FB-5A FB-56 K=2
Unaffected
Cleft subtype
Unaffected, CL, CP or CLP
FB-5A FB-56 K=4
1 cleft lip and palate
2 cleft lip
3 cleft palate
4 unaffected
Syndrome diagnosis
49 genetic syndromes
FB-TJ0 K=49
1 syndrome A
2 syndrome B
3 syndrome C
4 … to 49
Increasing diagnostic specificity
d Patient-level diagnostic prioritization
Confirmed diagnosis: Pitt-Hopkins syndrome
3 scans, one patient
PointNet++
z1
z2
z3
average
z̄
Patient-level facial embedding
Syndrome-level ranking
Output: ranked diagnostic candidates
vs. 49 syndrome prototypes
1 Pitt–Hopkins syndrome 0.87 ✓
2 Rett syndrome 0.67
3 Rubinstein–Taybi syndrome 0.66
4 Osteogenesis imperfecta 0.61
5 1p36 deletion 0.60
Patient-level retrieval
Output: similar reference cases
vs. 1,949 confirmed reference patients
#1 Pitt–Hopkins syndrome cos 0.90
#2 Pitt–Hopkins syndrome cos 0.89
#3 Pitt–Hopkins syndrome cos 0.88
Potential information for expert review
Ranked candidate syndromes
1 Pitt–Hopkins syndrome ✓
2 Rett syndrome
3 Rubinstein–Taybi syndrome
Similar confirmed cases
#1–#3 Pitt–Hopkins syndrome ✓
✓ confirmed diagnosis